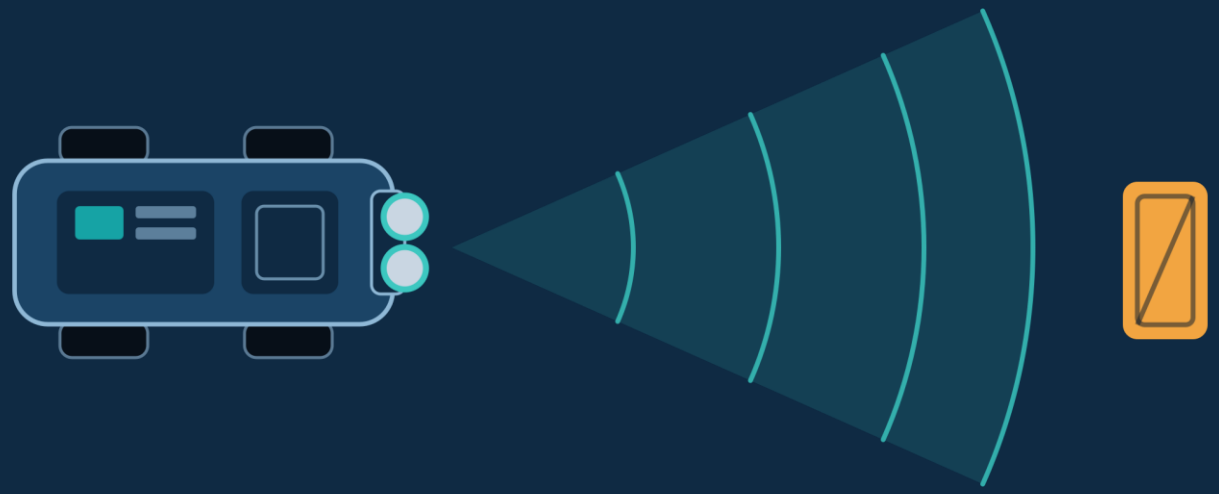
HORIZON 2.0 · BUILDATHON

# Team - HELIOS

**Problem Statement 2**

## Autonomous Mobility

Develop a vehicle that can move independently, detect obstacles, and navigate its surroundings without continuous human control.







**Team Leader – Satyam Tripathi**

Team Members:

Utkarsh Yadav ( AI&ML)

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Anant Vishwakarma (ECE)

Anvesha Dubey (CSE)

Ankit Sharma (AI&ML)

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Himani Goswami (AI&ML)

Shivani Maurya (EE)

Praduamna Shukla (ECE)

HELIOS

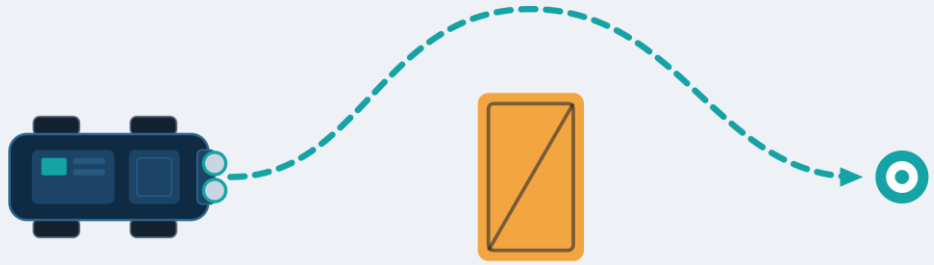
# ANV-4

Autonomous Vehicle -4

A low-cost vehicle that senses obstacles, plans a safe path and drives itself.

HORIZON 2.0 BUILDATHON

- **Obstacle Detection:** Uses **ultrasonic sensors** for real-time obstacle detection and continuous **surrounding monitoring**.
- **Intelligent Path Planning:** Computes the **shortest feasible, collision-free path** using real-time sensor data and navigation algorithms.
- **Dynamic Obstacle Avoidance:** Continuously detects obstacles and performs **real-time trajectory adjustments** for safe navigation.
- **Autonomous Navigation:** Enables **self-directed movement** through automated motor control with **minimal human intervention**.
- **Automatic Braking:** Activates an **emergency stop mechanism** when obstacles enter a predefined **critical safety distance**.
- **Low-Cost Architecture:** Integrates **affordable microcontrollers, ultrasonic sensors, and motor drivers** for cost-effective implementation.
- **Real-Time Decision Making:** Processes **live sensor inputs** to generate immediate **movement and direction commands**.



Move independently, detect obstacles and navigate without continuous human control.

Automaticly changes its paths after detecting the obsticle with its own Path planning navigation algorithm used.

HELIOS

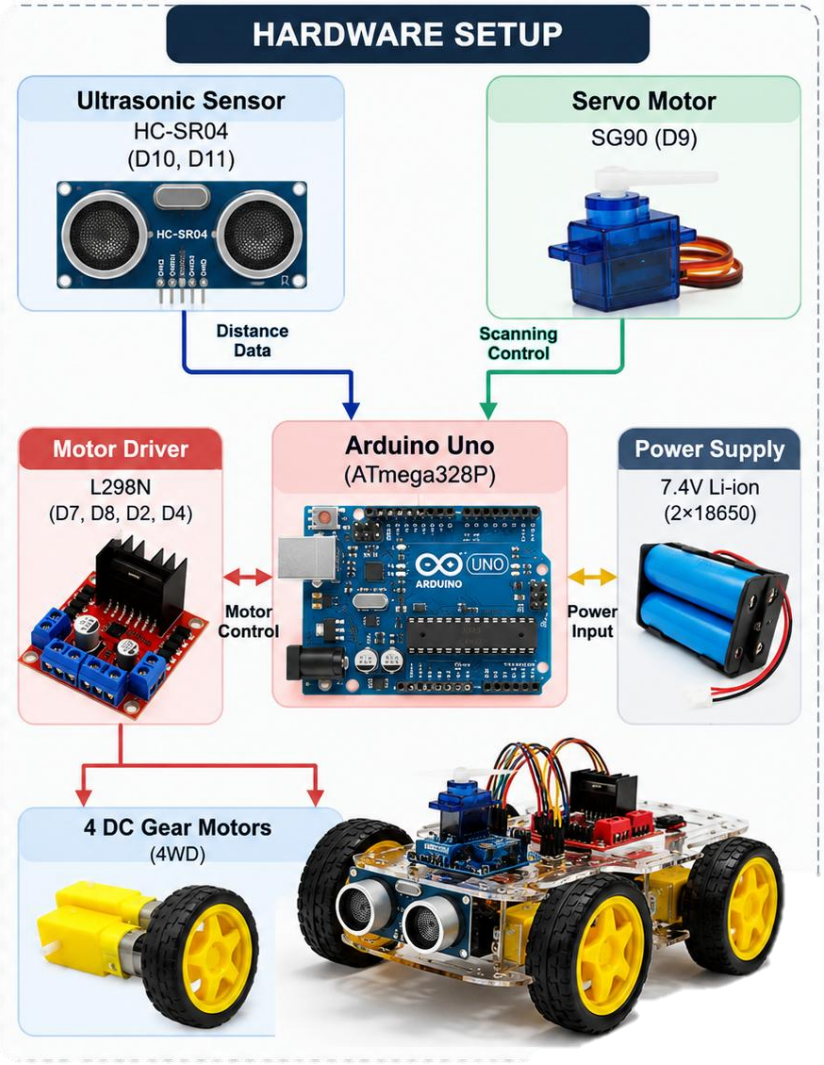
# TECHNICAL APPROACH

HORIZON 2.0
BUILDATHON

- **Programming Environment:** Arduino IDE with **Embedded C/C++** for hardware integration and autonomous control.
- **Microcontroller:** Arduino Uno (**ATmega328P**) processes sensor data and executes navigation commands.
- **Obstacle Detection:** HC-SR04 ultrasonic sensor uses **TRIG (D10)** and **ECHO (D11)** for real-time distance measurement.
- **Directional Scanning:** SG90 Servo (**D9**) rotates the sensor at **160° Left, 90° Center, and 20° Right**.
- **Motor Driver:** L298N controls **4 DC gear motors** using direction pins **D7, D8, D2, D4**.
- **Speed Regulation:** PWM signals through **ENA (D5)** and **ENB (D6)** regulate left and right motor speeds.
- **Decision Algorithm:** Implements **threshold-based obstacle avoidance** with a **20 cm safety distance**.

**Navigation Logic:** Executes **Move → Detect → Stop → Scan → Reverse/Turn**

**Power Supply:** **7.4V Li-ion battery pack (2×18650)** powers the 4WD mobility platform.



# FEASIBILITY AND VIABILITY



## FEASIBILITY

- Affordable Components: Uses low-cost sensors, microcontrollers, and motor drivers.
- Simple Implementation: Requires basic embedded programming and robotics principles.
- Easy Prototyping: Can be built using a compact wheeled chassis.
- Technical Feasibility: Uses readily available hardware and proven obstacle-avoidance techniques.

## VIABILITY

- Real-World Applications: Suitable for campus delivery and warehouse navigation.
- Scalable Design: Supports future upgrades with cameras and advanced navigation algorithms.
- Long-Term Potential: Offers a cost-effective foundation for autonomous mobility solutions.
- Commercial Viability: Can be adapted for indoor logistics and material transportation.






HELIOS

# IMPACT AND BENEFITS

HORIZON 2.0 BUILDATHON

## IMPACT

- Collision Reduction: Detects obstacles in real time.
- Improved Safety: Minimizes collision risks.
- Autonomous Mobility: Reduces human intervention.
- Efficient Navigation: Adapts to nearby obstacles.
- SDG 9: Promotes engineering innovation.

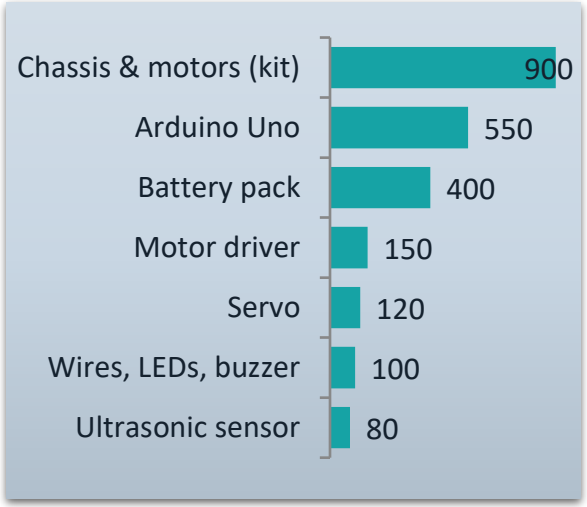
## BENEFITS

- Low-Cost Design: Uses affordable components.
- Easy Implementation: Simple hardware and programming.
- Versatile Use: Suitable for indoor environments.
- Scalable System: Supports future upgrades.
- Practical Application: Enables automated transportation




UN SDG CONNECTION

| | |
|---|---|
| SDG 9 | **Industry, Innovation & Infrastructure** Affordable autonomous-vehicle technology |
| SDG 11 | **Sustainable Cities & Communities** Safer, smarter mobility |
| SDG 3 | **Good Health & Well-being** Fewer collisions and injuries |



# RESEARCH AND REFERENCES



**1. Arduino UNO R3 — Microcontroller & Hardware Specifications**
https://docs.arduino.cc/hardware/uno-rev3/
**2. Arduino IDE — Programming Environment**
https://docs.arduino.cc/software/ide/
**3. HC-SR04 — Ultrasonic Distance Sensor Manual**
https://www.arobose.com/docs/HC-SR04/HC-SR04_Manual.pdf
**4. L298N — DC Motor Driver Documentation**
https://www.st.com/en/motor-drivers/l298.html
**5. Arduino Servo Library — Servo-Based Directional Scanning**
https://docs.arduino.cc/libraries/servo/
**6. United Nations — SDG 9: Industry, Innovation & Infrastructure**
https://sdgs.un.org/goals/goal9